



GIRRAWEE HIGH SCHOOL
YEAR 11 YEARLY EXAMINATION

2009

MATHEMATICS

2 UNIT

Time allowed- Two Hours
(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.
- Write your name on each piece of paper.

Question 1 (18 marks)**Marks**

(a) Find, correct to 3 decimal places: $\frac{\sqrt{421.6}}{17.93 + 4.9}$ 2

(b) Factorise fully:

(i) $5x^2 - 2x - 3$ (ii) $x^4 - y^4$ 4

(c) Simplify: $\frac{2}{3} - \frac{x-1}{4}$ 3

(d) Solve: $\frac{3}{x+2} = \frac{5}{2x-1}$ 3

(e) Solve using quadratic formula: $x^2 + 2 = -8x$ 3

(f) Find the value of the integers a, b and c if:

$\frac{6}{2\sqrt{7}-5} = a + b\sqrt{c}$ 3

Question 2 (21 marks)

(a) Describe (write the equation and name the geometrical shape) the locus of the following: 3

(i) a point that moves so that it is always 2 units above the x axis.

(ii) a point that moves so that it is always 5 units from the point $(1, -2)$.

(b) Find the equation of the locus of a point P that moves so that PA is twice the distance of PB where $A = (0, 3)$ and $B = (4, 7)$. 4

(c) Find the equation of the locus of a point which moves in a plane so that its distance from the point $S(-2, 4)$ is equal to its distance from the line $y = 6$. 4

(d) For the following parabolas find 10

(i) the coordinates of the focus.

(ii) the equation of the directrix.

(iii) Sketch the parabola showing the above features.

(α) $y^2 = 6x$ (β) $x^2 = -8y$

Question 3 (16 marks)

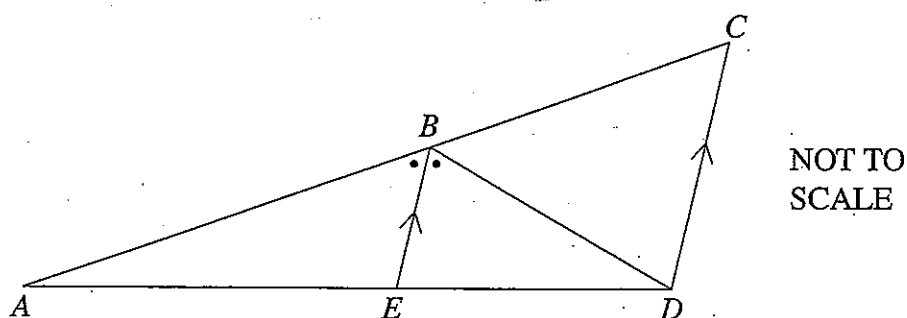
(a) A regular polygon has 9 sides. Find

2

(i) the angle sum.

(ii) the size of each interior angle.

(b) In the diagram, $BE \parallel CD$ and BE bisects $\angle ABD$.



Copy the diagram into your writing booklet.

(i) Explain why $\angle EBD = \angle BDC$.

2

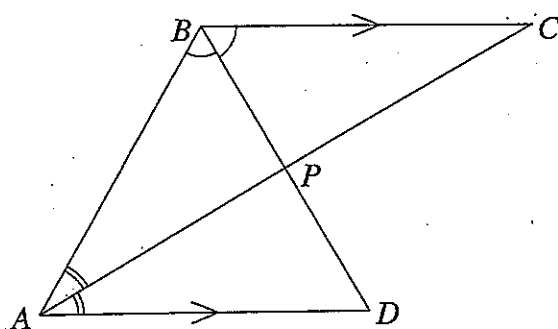
(ii) Prove that $\triangle BCD$ is isosceles.

2

(iii) Hence show that $AE : ED = AB : BD$

2

(c) In the diagram below AD is parallel to BC , AC bisects $\angle BAD$ and BD bisects $\angle ABC$. The lines AC and BD intersect at P .



Copy the diagram into your writing booklet.

(i) Prove that $\angle BAC = \angle BCA$.

2

(ii) Prove that $\triangle APB \equiv \triangle CBP$.

3

(iii) Prove that $ABCD$ is a rhombus.

3

Question 4 (30 marks)

(a) For each of the following curves

$(\alpha) y = -x^2 - 3$

$(\beta) y = |x + 1|$

$(\gamma) y = \frac{1}{x+2}$

$(\delta) y = \sqrt{7+x}$

16

(i) Draw a neat sketch showing important features.

(ii) State the domain and range.

(b) Find the domain of $y = \frac{1}{\sqrt{64-x^2}}$. 2(c) Is the function $f(x) = x^4 + 2x$ odd, even or neither? Justify your answer. 3

(d) A function is defined by the rule

$$g(x) = \begin{cases} x^2 - 4 & x \geq -2 \\ x + 2 & x < -2 \end{cases}$$

(i) Sketch the curve. 3

(ii) Find the value of $g(2)$, and $g(-3)$. 2(e) Shade the region given by $x^2 + y^2 \leq 9$ and $x - y > 2$. 4**Question 5** (23 marks)

(a) Differentiate: 11

$(i) y = 4x^2 - 7x + 3 + 5x^{-2}$

$(ii) y = (2x + 5)(x^2 - 1)^4$

$(iii) y = \frac{5x}{x+1}$

$(iv) y = \sqrt{2x-7}$

(b) Evaluate the following limits: 4

$(i) \lim_{x \rightarrow 4} \frac{x-4}{x^2-16}$

$(ii) \lim_{x \rightarrow \infty} \frac{4x^2+1}{3x^2}$

(c) If α and β are the roots of the quadratic equation $3y^2 - 8y + 3 = 0$, find the value of

$(i) \alpha + \beta \quad (ii) \alpha\beta \quad (iii) \alpha^2 + \beta^2$ 4

(d) For what values of k is the expression $k(x+1)(x-8) - (x-4)$ negative definite? 4

Question 6 (12 marks)

- (a) Find the perpendicular distance from
- $(-2, 3)$
- to the line

$$5x - 12y + 20 = 0.$$

2

- (b) (i) Find the equation of the straight line
- l
- through
- $(-1, 2)$
- that is perpendicular to the line
- $3x + 6y - 7 = 0$
- .

3

- (ii) Line
- l
- cuts the
- x
- axis at
- P
- and
- y
- axis at
- Q
- . Find the coordinates of
- P
- and
- Q
- .

2

- (iii) Find the area of
- $\triangle OPQ$
- where
- O
- is the origin.

2

- (c) Show that the lines
- $x - 5y - 17 = 0$
- ,
- $3x - 2y - 12 = 0$
- and
- $5x + y - 7 = 0$
- are concurrent.

3

Question 7 (16 marks)

- (a) Solve for
- $0^\circ \leq \theta \leq 360^\circ$
- . Write answer correct to the nearest minute.

5

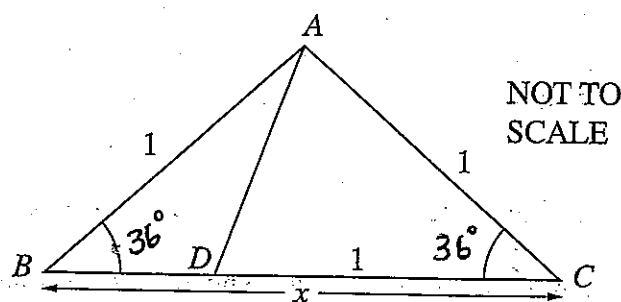
(i) $\cos \theta = \frac{-1}{\sqrt{2}}$

(ii) $\tan 2\theta = \sqrt{3}$

- (b) Prove:
- $(\sin \theta + \cos \theta)^2 - (\sin \theta - \cos \theta)^2 = 4 \sin \theta \cos \theta$

3

- (c) In the diagram,
- ABC
- is an isosceles triangle where
- $\angle BAC = 108^\circ$
- and
- $AB = AC = 1$
- . The point
- D
- is chosen on
- BC
- such that
- $CD = 1$
- . Let
- $BC = x$
- .



- (i) Show that
- $\angle ADC = 72^\circ$
- and hence show that triangles
- DBA
- and
- ABC
- are similar.

3

- (ii) Hence deduce that
- $x^2 - x - 1 = 0$
- .

2

- (iii) By using the cosine rule, deduce that
- $\cos 36^\circ = \frac{1 + \sqrt{5}}{4}$

3

Year 11 Mathematics Yearly 2009 - solutions.

Question 1 (18 marks)

(a) 0.899 (1)

(b) (i) $5x^2 - 2x - 3$

$(-5, 3)$ $p+q = -2$

$$= 5x^2 - 5x + 3x - 3$$

$$= 5x(x-1) + 3(x-1)$$

$$= (x-1)(5x+3) \quad (2)$$

(ii) $x^4 - y^4$

$$= (x^2)^2 - (y^2)^2$$

$$= (x^2 + y^2)(x^2 - y^2) \quad (3)$$

$$= (x^2 + y^2)(x+y)(x-y)$$

(c) $\frac{2}{3} - \frac{x-1}{4}$

$$= \frac{8 - 3(x-1)}{12} \quad (3)$$

$$= \frac{8 - 3x + 3}{12} = \frac{11 - 3x}{12}$$

(d) $\frac{3}{x+2} = \frac{5}{2x-1}$

$$3(2x-1) = 5(x+2)$$

$$6x - 3 = 5x + 10$$

$$6x - 5x = 13 \quad (3)$$

$$x = 13$$

(e) $x^2 + 2 = -8x$

$$x^2 + 8x + 2 = 0$$

$$x = \frac{-8 \pm \sqrt{64 - 4 \times 1 \times 2}}{2}$$

$$= \frac{-8 \pm \sqrt{64 - 8}}{2} = \frac{-8 \pm \sqrt{56}}{2}$$

$$= \frac{-8 \pm \sqrt{14 \times 4}}{2} = \frac{-8 \pm 2\sqrt{14}}{2} \quad (3)$$

$$= \frac{2(-4 \pm \sqrt{14})}{2} = \underline{\underline{-4 \pm \sqrt{14}}}$$

(f) $\frac{6}{2\sqrt{7}-5} = \frac{6}{2\sqrt{7}-5} \times \frac{2\sqrt{7}+5}{2\sqrt{7}+5}$

$$= \frac{6(2\sqrt{7}+5)}{(2\sqrt{7}-5)(2\sqrt{7}+5)}$$

$$= \frac{12\sqrt{7} + 30}{(2\sqrt{7})^2 - 5^2} = \frac{12\sqrt{7} + 30}{28 - 25}$$

$$= \frac{12\sqrt{7} + 30}{3}$$

$$= 4\sqrt{7} + 10 \quad (3)$$

$$= 10 + 4\sqrt{7}$$

$$\underline{\underline{a = 10, b = 4, c = 7}}$$

Question 2 (21 marks)

Page 2

(a) (i) $y=2$, straight line

(1)

(ii) $(x-1)^2 + (y+2)^2 = 25$, circle

(2)

(b) $P(x, y)$ $A(0, 3)$ $B(4, 7)$

$$PA = 2 \times PB$$

$$\sqrt{x^2 + (y-3)^2} = 2 \times \sqrt{(x-4)^2 + (y-7)^2}$$

$$x^2 + (y-3)^2 = 4[(x-4)^2 + (y-7)^2] \quad (4)$$

$$x^2 + y^2 - 6y + 9 = 4[x^2 - 8x + 16 + y^2 - 14y + 49]$$

$$x^2 + y^2 - 6y + 9 = 4x^2 - 32x + 64 + 4y^2 - 56y + 196$$

$$4x^2 - x^2 + 4y^2 - y^2 - 32x - 56y + 260 - 9 = 0$$

$$\underline{\underline{3x^2 + 3y^2 - 32x - 56y + 251 = 0}}$$

(c) Let $P(x, y)$ be the variable point. Let M be the foot of the perpendicular from P to the line $y=6$

Given that $PS = PM$

$P(x, y)$ $S(-2, 4)$

$$\sqrt{(x+2)^2 + (y-4)^2} = |y-6| \quad (4)$$

$$(x+2)^2 + (y-4)^2 = |y-6|^2 = (y-6)^2 \quad (\because |x|^2 = x^2)$$

$$x^2 + 4x + 4 + y^2 - 8y + 16 = y^2 - 12y + 36$$

$$x^2 + 4x = -4y + 16$$

$$\underline{\underline{(x+2)^2 = -4(y-5)}}$$

(d) (a) (i) $y^2 = 6x$

$$y^2 = 4ax$$

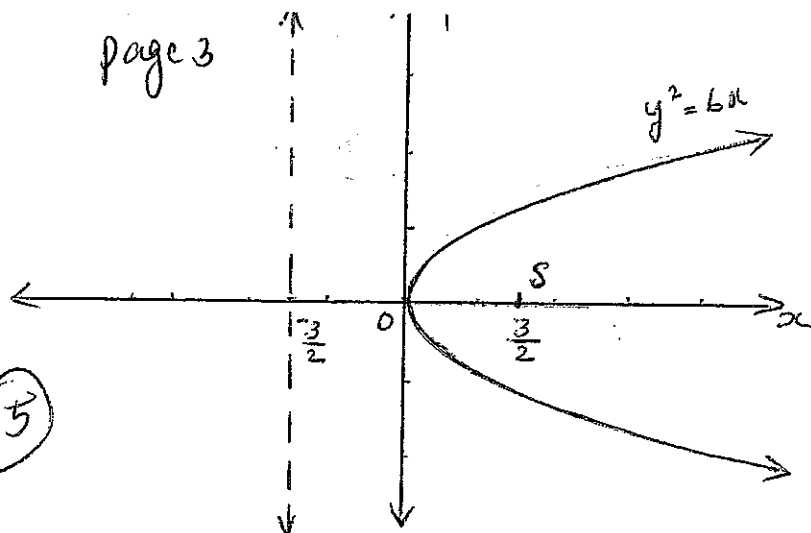
$$4a = 6; a = \frac{3}{2}$$

$$\text{Focus} = \left(\frac{3}{2}, 0\right)$$

$$\text{directrix: } x = -\frac{3}{2}$$

(5)

page 3



(b) $x^2 = -8y$

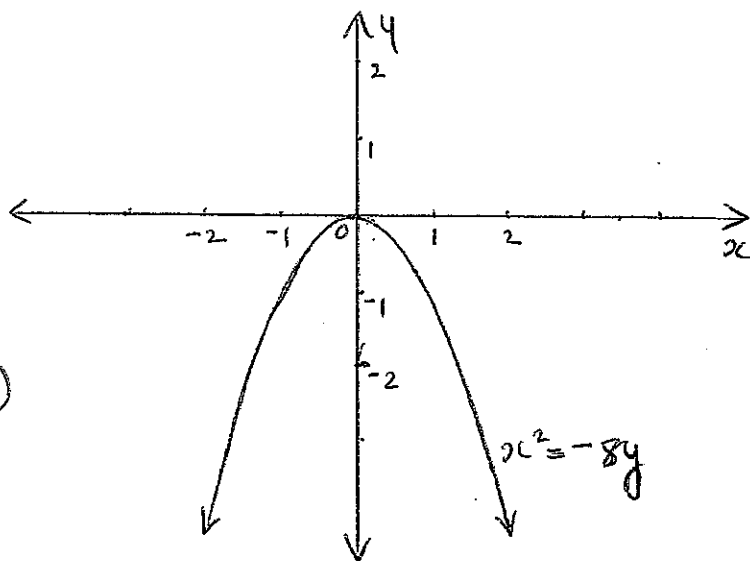
$$x^2 = -4ay$$

$$4a = 8; a = 2$$

$$\text{Focus } (0, -2)$$

$$\text{directrix: } y = 2$$

(5)



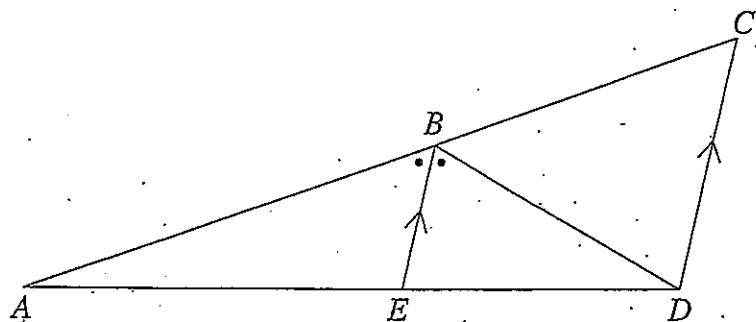
Question 3 (16 marks)

(a) (i) Angle sum $= (n-2) \times 180$

$$= (9-2) \times 180 = \underline{\underline{1260^\circ}} \quad (1)$$

(ii) One interior angle $= \frac{1260}{9} = \underline{\underline{140^\circ}} \quad (1)$

(b)



(i) $\angle EBD = \angle BDC$ (alternate angles equal, $BE \parallel CD$)

(2)

page 4

(ii) $\angle ABE = \angle BCD$ (Corresponding angles equal, $BE \parallel CD$)

$$\angle EBD = \angle BDC \text{ (from (i))}$$

$$\angle EBD = \angle ABE \text{ (given)}$$

(2)

$$\therefore \angle BCD = \angle BDC$$

$\therefore \triangle BCD$ is isosceles (two equal angles)

(iii) $\frac{AE}{ED} = \frac{AB}{BC}$ (equal ratios on transversals by parallel lines)

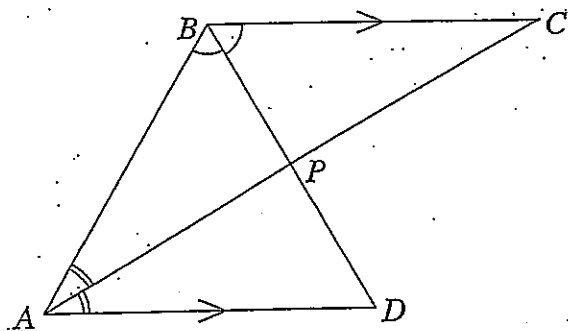
$$BC = BD \text{ (isosceles } \triangle BCD)$$

$$\therefore \frac{AE}{ED} = \frac{AB}{BD}$$

(2)

$$\text{ie } \underline{\underline{AE:ED = AB:BD}}$$

(c)



(i) $\angle BAC = \angle CAD$ (AC bisects $\angle BAD$)

$\angle CAD = \angle BCA$ (alternate angles equal, $BC \parallel AD$)

(2)

$$\therefore \angle BAC = \angle BCA$$

(ii) Consider $\triangle ABP$ and $\triangle CBP$

$$\angle BAC = \angle BCA \text{ (from (i))}$$

$$\angle ABP = \angle CBP \text{ (BP bisects } \angle ABC)$$

BP is common

$$\therefore \underline{\underline{\triangle ABP \equiv \triangle CBP \text{ (AAS)}}}$$

(3)

(iii) $AP = CP$ (matching sides of congruent triangles ABP and CBP)
 $\angle APB = \angle CPB$ (matching angles of congruent triangles ABP and CBP)

$\angle APB + \angle CPB = 180^\circ$ (APC is a straight angle, 180°)

$$\therefore \angle APB = \angle CPB = 90^\circ$$

In $\triangle ABP$ and $\triangle ADP$

$\angle BAP = \angle DAP$ (AP bisects $\angle BAD$)

$\angle APB = \angle APD$ ($BP \perp AC$)

AP is common

(3)

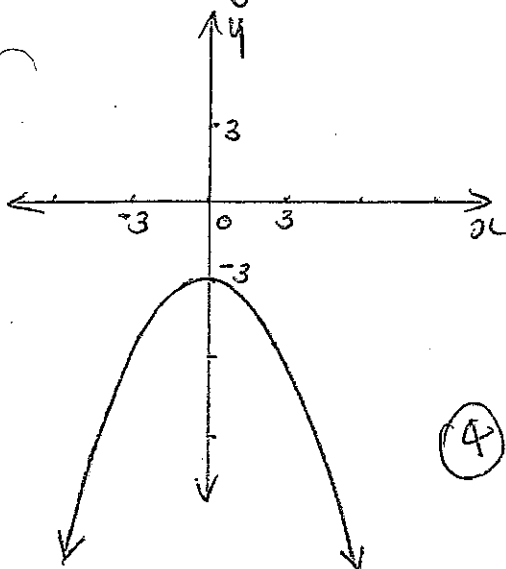
$\therefore \triangle ABP \equiv \triangle ADP$ (AAS)

$BP = PD$ (matching sides of congruent triangles ABP and ADP)

$ABCD$ is a rhombus (diagonals bisect each other at right angles)

Question 4 (30 marks)

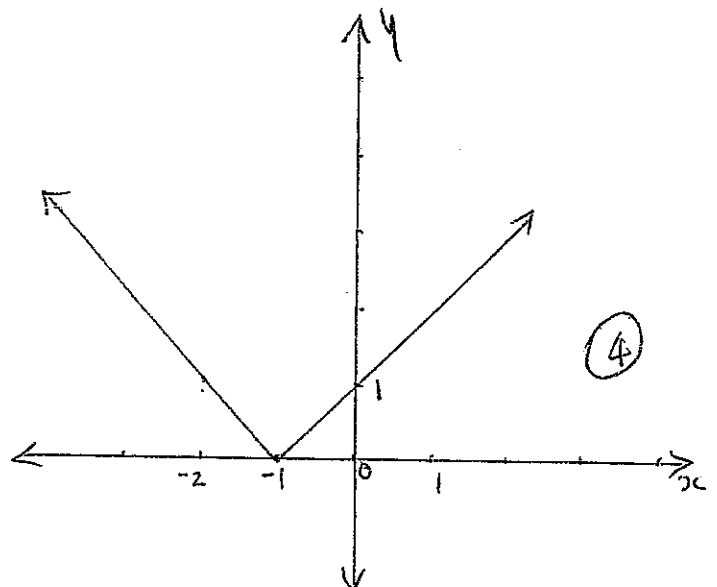
(a) (a) $y = -x^2 - 3$



D : all real x

R : $y \leq -3$, y real

(b) $y = |x + 1|$



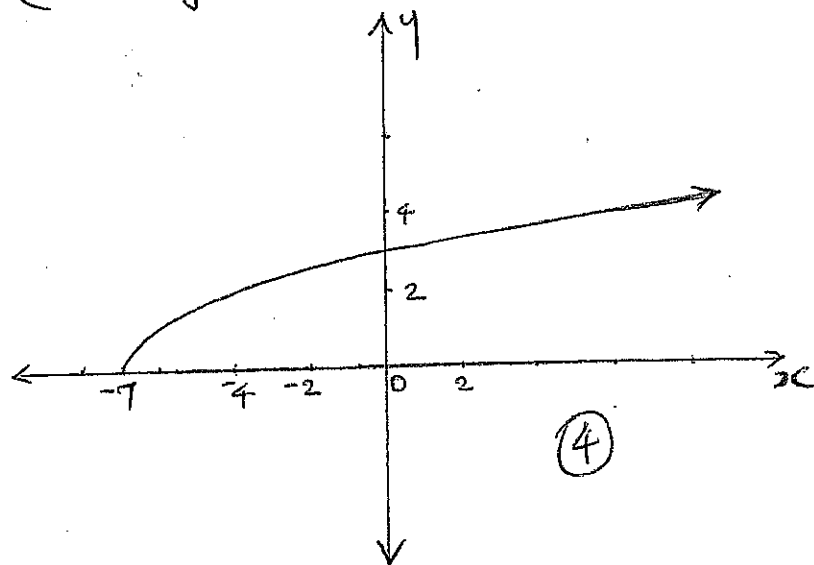
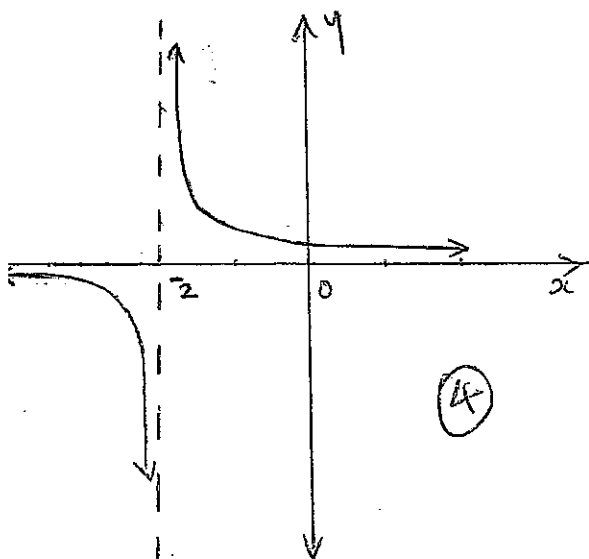
D : all real x

R : $y \geq 0$, y real

$$(r) \quad y = \frac{1}{x+2}$$

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$$(d) \quad y = \sqrt{7+x}$$



D: all real x , $x \neq -2$

R: all real y , $y \neq 0$

D: $x \geq -7$

R: $y \geq 0$

$$(b) \quad y = \frac{1}{\sqrt{64-x^2}}$$

$\sqrt{64-x^2}$ exists when

$$64 - x^2 \geq 0$$

$$64 \geq x^2$$

$$x^2 \leq 64$$

$$-8 \leq x \leq 8$$

$$D: \underline{-8 \leq x \leq 8}$$

$$(c) \quad f(x) = x^4 + 2x$$

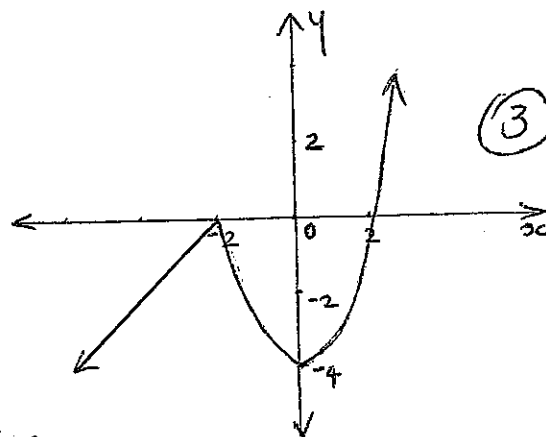
$$f(-x) = (-x)^4 + 2(-x) \\ = x^4 - 2x$$

$$f(-x) \neq f(x)$$

$$\text{and } f(-x) \neq -f(x)$$

$\therefore f(x)$ is neither even nor odd function.

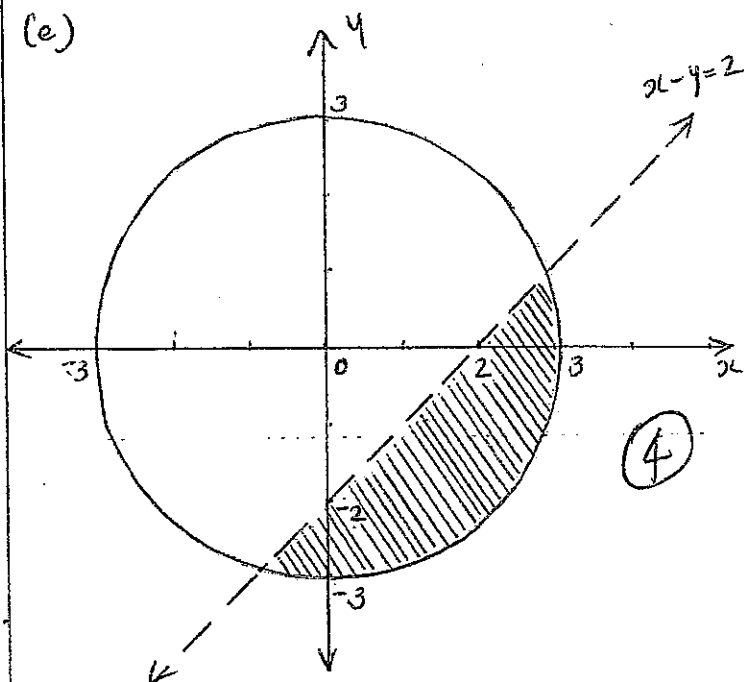
(d)



$$g(2) = 0$$

$$g(-3) = -3 + 2 = -1$$

(e)



Question 5 (23 marks)

(a) (i) $y = 4x^2 - 7x + 3 + 5x^{-2}$

$$y' = 4 \times 2x - 7 + 5 \times -2x^{-3}$$

$$= \underline{\underline{8x - 7 - 10x^{-3}}} \quad (2)$$

(iii) $y = \frac{5x}{x+1}$

$$y' = \frac{(x+1) \times 5 - 5x \times 1}{(x+1)^2}$$

$$= \frac{5x + 5 - 5x}{(x+1)^2}$$

$$= \underline{\underline{\frac{5}{(x+1)^2}}} \quad (3)$$

(b) (i) $\lim_{x \rightarrow 4} \frac{x-4}{x^2-16} = \lim_{x \rightarrow 4} \frac{x-4}{(x+4)(x-4)}$

$$= \lim_{x \rightarrow 4} \frac{1}{x+4}$$

$$= \underline{\underline{\frac{1}{8}}} \quad (2)$$

page 7

(ii) $y = (2x+5)(x^2-1)^4$

$$y' = (2x+5) + (x^2-1)^3 \times 2x$$

$$+ (x^2-1)^4 \times 2 \quad (3)$$

$$= \underline{\underline{8x(2x+5)(x^2-1)^3 + 2(x^2-1)^4}}$$

(iv) $y = \sqrt{2x-7}$

$$y' = \frac{1}{2\sqrt{2x-7}} \times 2$$

$$= \underline{\underline{\frac{1}{\sqrt{2x-7}}}} \quad (3)$$

(ii) $\lim_{x \rightarrow \infty} \frac{4x^2+1}{3x^2}$

$$= \lim_{x \rightarrow \infty} \frac{4 + \frac{1}{x^2}}{3} \quad (2)$$

$$= \underline{\underline{\frac{4}{3}}} \quad \left(\because \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \right)$$

(c) $3y^2 - 8y + 3 = 0$

(i) $\alpha + \beta = -\frac{(-8)}{3} = \underline{\underline{\frac{8}{3}}} \quad (1)$

(ii) $\alpha\beta = \frac{3}{3} = \underline{\underline{1}} \quad (1)$

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(\frac{8}{3}\right)^2 - 2 = \frac{64}{9} - 2 = \underline{\underline{5\frac{1}{9}}} \quad (2)$$

$$(d) k(x+1)(x-8) - (x+4)$$

$$= k(x^2 - 7x - 8) - x - 4$$

$$= kx^2 - 7kx - 8k - x - 4$$

$$= kx^2 - x(7k+1) - 4 - 8k \quad \text{--- (1)}$$

(1) is negative definite

$$\Rightarrow k < 0 \text{ and } \Delta < 0$$

$$\Delta = (7k+1)^2 - 4k(-4-8k)$$

$$= 49k^2 + 14k + 1 + 16k + 32k^2$$

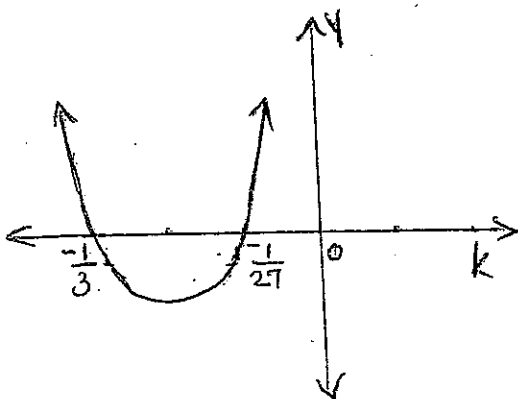
$$= 81k^2 + 30k + 1$$

X intercepts of Δ are given by

$$k = \frac{-30 \pm \sqrt{900 - 4 \times 81 \times 1}}{2 \times 81}$$

$$= \frac{-30 \pm \sqrt{900 - 324}}{162}$$

$$= -\frac{1}{27} \text{ or } -\frac{1}{3}$$



From the graph $\Delta < 0$

$$\text{when } -\frac{1}{3} < k < -\frac{1}{27}$$

Question 6 (12 marks)

$$(a) d = \frac{|5x - 2 - 12 \times 3 + 20|}{\sqrt{5^2 + 12^2}}$$

$$= \frac{|-10 - 36 + 20|}{13}$$

$$= \frac{26}{13} = 2 \quad \text{(2)}$$

$$(b)(i) 3x + 6y - 7 = 0$$

$$6y = -3x + 7$$

$$y = -\frac{3x}{6} + \frac{7}{6}$$

$$\text{gradient} = -\frac{1}{2}$$

$$\text{gradient of } l = 2$$

Equation of l

$$y - 2 = 2(x + 1) \quad \text{(3)}$$

$$y - 2 = 2x + 2$$

$$\underline{\underline{y = 2x + 4}}$$

$$(ii) y = 2x + 4$$

$$y = 0 \Rightarrow 2x + 4 = 0$$

$$2x = -4 \quad x = -\frac{4}{2} = -2$$

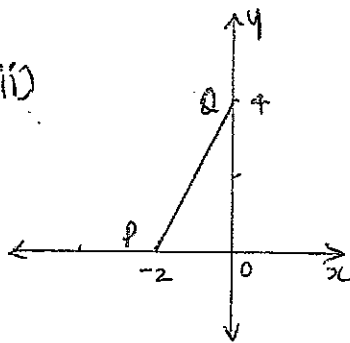
$$\underline{\underline{P(-2, 0)}}$$

$$x = 0 \Rightarrow y = 4$$

$$\underline{\underline{Q = (0, 4)}}$$

(2)

(iii)

Area of $\triangle OPQ$

(2)

$$= \frac{1}{2} \times 2 \times 4$$

$$= \underline{\underline{4 \text{ unit}^2}}$$

$$(C) \quad x - 5y = 17 \quad \text{--- (1)}$$

$$\bigcirc \quad 3x - 2y = 12 \quad \text{--- (2)}$$

$$\textcircled{1} \times 3$$

$$3x - 15y = 51 \quad \text{--- (3)}$$

$$3x - 2y = 12 \quad \text{--- (2)}$$

$$\textcircled{3} - \textcircled{2} \quad -13y = 39$$

$$y = -3$$

$$x = 17 + 5y$$

$$= 17 - 15 = 2$$

Point of intersection of

① and ② is $(2, -3)$ Substitute $(2, -3)$ in

$$5x + y - 7 = 0$$

$$\text{LHS} = 5 \times 2 + (-3) - 7$$

$$= 10 - 3 - 7 = 0$$

$$\text{RHS} = 0$$

$$\text{LHS} = \text{RHS}$$

(3)

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The point $(2, -3)$ is on the line $5x + y - 7 = 0$. Therefore the given lines are concurrent.

Question 7 (16 marks)

$$(a) (i) \cos \theta = \frac{-1}{\sqrt{2}}$$

$$\text{acute } \angle \theta = 45^\circ$$

$$\theta = 180 - 45, 180 + 45 \quad \textcircled{2}$$

$$= \underline{\underline{135^\circ, 225^\circ}}$$

$$(ii) \tan 2\theta = \sqrt{3}, \quad 0 \leq \theta \leq 360^\circ$$

$$\text{Let } u = 2\theta \quad 0 \leq 2\theta \leq 720^\circ$$

$$\tan u = \sqrt{3}, \quad 0 \leq u \leq 720^\circ$$

$$u = 60^\circ, 240^\circ, 420^\circ, 600^\circ \quad \textcircled{3}$$

$$\theta = \underline{\underline{30^\circ, 120^\circ, 210^\circ, 300^\circ}}$$

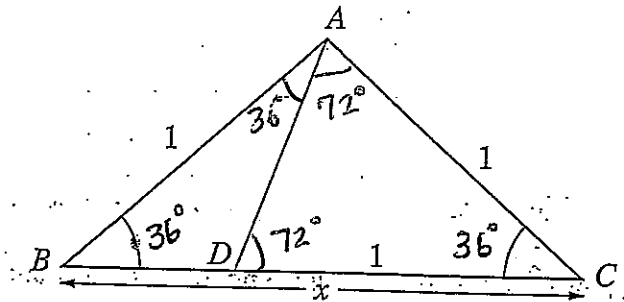
(b) LHS = $(\sin\theta + \cos\theta)^2 - (\sin\theta - \cos\theta)^2$ page 10

$$= \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta - (\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta)$$

$$= 1 + 2\sin\theta\cos\theta - 1 + 2\sin\theta\cos\theta \quad (3)$$

$$= 4\sin\theta\cos\theta = \text{RHS.}$$

(c)



(i) $\angle ADC = \frac{180 - 36}{2} = 72^\circ$ ($\because \triangle ADC$ is isosceles)

$$\angle ADC = \angle DBA + \angle BAD \text{ (exterior angle theorem)}$$

$$\text{i.e. } 72 = 36 + \angle BAD$$

$$\therefore \angle BAD = 36^\circ$$

In $\triangle DBA$ and $\triangle ABC$

$$\angle DBA = \angle ACB = 36^\circ \quad (3)$$

$$\angle BAD = \angle ABC = 36^\circ$$

$\therefore \triangle DBA \sim \triangle ABC$ (equiangular)

(ii) Since $\triangle DBA \sim \triangle ABC$ we have

$$\frac{AB}{BC} = \frac{BD}{AB} = \frac{DA}{AC}$$

$$\frac{1}{x} = \frac{x-1}{1} \quad (2)$$

$$1 = x^2 - x \quad \therefore x^2 - x - 1 = 0$$

(iii) In $\triangle ABC$,

page 11

$$\cos 36^\circ = \frac{x^2 + 1^2 - 1^2}{2 \cdot x \cdot 1}$$

$$\cos 36^\circ = \frac{x^2}{2x} = \frac{x}{2}$$

From (ii) we have $x^2 - x - 1 = 0$

$$x = \frac{-(-1) \pm \sqrt{1 - 4 \times 1 \times -1}}{2}$$

$$= \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

$$\therefore \cos 36^\circ = \frac{1 \pm \sqrt{5}}{2} \times \frac{1}{2} \quad (3)$$

$$= \frac{1 \pm \sqrt{5}}{4}$$

Since $\cos 36^\circ$ is positive we have

$$\cos 36^\circ = \frac{1 + \sqrt{5}}{4}$$

